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RESEARCH ARTICLE

# A Dual Perspective on Community Mobility: Home-Centric Place Networks and Older Adults' Mental Health in Shanghai During the Pandemic

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## ABSTRACT

Older adults' community mobility reflects both spatial and relational aspects of aging in place within urban environments. To capture this duality, we introduce a home-centric place network framework that links the home to community destinations older adults routinely visit. The spatial dimension is measured using standard movement metrics, while the relational dimension is defined as opportunities for social encounters (sociability) and operationalized as the overlap of neighbors' routine activities across community places. We tested this framework by collecting two waves of GIS-based mobility data and an online survey from 64 community-dwelling older adults in Shanghai, a rapidly aging megacity, during the COVID-19 pandemic. Results showed that older adults' place networks were diverse, but both the frequency of out-of-home visits and sociability declined sharply following a major outbreak. Those who were more mobile and sociable before the outbreak experienced greater reductions, and sociability was associated with more negative expectations. As social places became associated with higher infection risk, the pandemic fundamentally altered the relational meaning of community mobility. Overall, our findings demonstrate that a home-centric place network framework deepens population geography by linking physical mobility, social encounters, and the emotional well-being of older adults.

## 1 | Introduction

Aging in place underscores the importance of homes and local communities for older adults' mental health (Andrews and Phillips 2005). Recently, "place" has been re-conceptualized through a mobility lens: a local community is not a container but a set of physical movements in which everyday life unfolds (Sheller and Urry 2006; Sheller 2017). Geographers led this shift, emphasizing older adults' agency in navigating their environments (e.g., Artamonova et al. 2020; Bo et al. 2024), while social gerontologists have shown how social networks shape lived experience and well-being (Fiori et al. 2007; Russell et al. 2019). Although both fields increasingly recognize that spatial and relational dimensions are deeply intertwined (Webber et al. 2010; Yarker et al. 2024), their insights often remain parallel, limiting our understanding of how their interplay affects older adults' mental health. This gap is especially significant given mobility inequalities shaped by institutional, environmental, and socioeconomic factors (Hannam et al. 2006; Finlay and Finn 2020).

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Elaine Ho's "aging in networks" (Ho et al. 2024) responds to this need for synthesis by offering a conceptual framework that integrates spatial and relational dimensions. It highlights that older adults' mobility is not simply about physical movement, but is also a process of place-making rooted in daily social interactions. Drawing on both mobility and aging studies, this concept shows that mobility is closely linked to the network maintenance (Ho and Chiu 2020) and expansion (Gao et al. 2024; Matysiak 2022). This synthesis invites a new analytical agenda: to operationalize how aging unfolds within webs of social connection across diverse places—local, transnational, and virtual (see this special issue).

Building on this conceptual synthesis, we argue for an analytical approach that operationalizes both the spatial and relational dimensions of community mobility. While existing research demonstrates the significance of both dimensions for older adults' mental health—the spatial dimension reflecting agency and independence (Bevilacqua et al. 2024; Feng et al. 2024) and the relational dimension capturing active community life (Fiori et al. 2007; Russell et al. 2019)—these have largely been measured and analyzed in isolation. This integration is important, as for example, older adults' access to emotional and practical support is shaped not only by who is in their social network, but also by where those contacts are located and how easy it is to reach them. Our study brings the two strands into dialog by developing a home-centric place network, enabling us to analyze how spatial and relational dimensions jointly shape mental health outcomes.

The COVID-19 pandemic provides a critical case for applying this synthetic approach to community mobility and its implications for mental health. As a crisis that drastically limited both physical movement and daily social interactions (Ho et al. 2023), the pandemic exposed and intensified existing disparities among older adults. Evidence indicates that older adults experienced not only a profound contraction in daily mobility patterns (Choe et al. 2022), but also heightened levels of loneliness and anxiety (Asante et al. 2021; Litwin and Levinsky 2022). Integrating both dimensions allows for a nuanced understanding of how such disruptions intersect to influence mental health by identifying those who were most vulnerable, how inequalities manifested, and the mental health consequences of disrupted mobility and social connections.

Specifically, this study develops a home-centric place network to capture and integrate both spatial and relational aspects of community mobility. In the home-centric place network design, home is considered the "ego," locations visited in community mobility are "alters," and the frequency of visits to these locations is the "tie strength." The spatial dimension is operationalized as distance, frequency, and quantity, standard measures in the multidimensional mobility scale (Fillekes et al. 2019), and the relational dimension as opportunities for social encounters in community environments (Feld 1981).

Data were collected from two waves of GIS-based tracking and one self-reported survey with 64 older adults in Shanghai—first in November 2021 and again in September 2022, reflecting mobility behaviors before and after the city's major 2-month outbreak (April–June 2022). Our analysis addresses three questions: (1) To what extent did older adults have diverse home-centric place networks across both spatial and relational dimensions? (2) How did these networks change during the pandemic? (3) What associations can be observed between these changes and older adults' mental health and well-being, particularly in relation to emergent inequalities?

This study offers several contributions. First, it operationalizes the concept of "aging in networks" into a synthetic analytical framework that integrates spatial and relational dimensions of community mobility. This analytical framework aims to move beyond traditional notions of "spatial exposure" to conceptualize mobility as participation in relationally embedded spatial networks. Second, it brings the two literature strands into empirical dialog by using home-centric place networks to reveal patterns of diversity and vulnerability that single-dimension approaches miss, particularly for older adults' mental health. Third, it generates nuanced understanding about how intersecting spatial and relational disruptions during COVID-19 impact older adults' mental health. This dual perspective provides valuable insights for population geography by illuminating how the spatial and relational fabric of older adults' everyday lives is structured—and how crises can rapidly reshape these networks, with direct consequences for mental health.

## 1.1 | Literature Review

### *1.1.1 | Community Mobility: Spatial and Relational*

Community mobility captures the places in which aging unfolds in everyday life (Andrews and Phillips 2005; Webber et al. 2010). It encompasses two interrelated dimensions of aging experience. First, its spatial dimension reflects older adults' ability and opportunity to walk, navigate, and carry out daily activities in community settings (Feng et al. 2024). Second, its relational dimension concerns the social practices that sustain active community life, such as meeting friends and neighbors, volunteering, and participating in local activities (Yarker et al. 2024).

Nevertheless, the two dimensions of community mobility are often analyzed separately. On the spatial side, geographers draw on fine-scale, real-time GIS records to capture older adults' activity space, the set of locations individuals routinely visit (Feng et al. 2024). They also develop a range of mobility indicators (Fillekes et al. 2019) to assess how spatial configurations of movement relate to physical and mental health (Klein et al. 2023; Crane et al. 2023). For example, Crane et al. (2023) show that larger activity-space elongation, greater time spent outside the home, and longer travel distances are associated with improved physical fitness.

On the relational side, social gerontologists long emphasized the significance of social ties for healthy aging. Aging in place is about maintaining existing social networks and staying connected to the community (Yarker et al. 2024). According to social convoy theory, older adults need various sources of social support to thrive, with neighbors playing a vital role in providing daily help and companionship alongside family members (Russell et al. 2019; Cornwell et al. 2008).



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In recent years, scholars from both lines of research have recognized the co-constitutive relationship between the spatial and relational dimensions of community mobility. Geographical studies show that mobility is indispensable for older adults to maintain key social networks, particularly with children and relatives (Artamonova et al. 2020; Zilincikova et al. 2024). Sociologists likewise demonstrate that many social interactions in communities are not deliberately planned; rather, they emerge through routine encounters as individuals go about their everyday activities (Browning et al. 2017).

To integrate these insights, Ho et al. (2024) propose the concept of "aging in networks," which integrates both spatial and relational dimensions of older adults' mobility. This concept directs attention to a variety of places where social networks are formed, maintained, and expanded. In fact, older adults can adopt very diverse mobility patterns, from local communities surrounding one's home (Gardner 2011), to other cities (Bo et al. 2024; Gillespie and Fokkema 2024) or to other countries (King et al. 2019; Sampaio 2020). These mobile older adults can maintain their existing networks (Ho and Chiu 2020) and form new connections at other destinations (Gao et al. 2024; Matysiak 2022). Thus, the aging-in-networks view older adults' mobility as social interactions that occur through physical movements.

Despite this growing effort in theoretical synthesis, few studies have yet provided a synthetic analytical framework to articulate this dual perspective on community mobility for older adults. Most current research either uses qualitative GIS approaches (e.g., Gardner 2011) or relies on survey data (e.g., Song et al. 2023; Lane et al. 2020). An exception is Franke et al. (2017), who combined GIS, surveys, and interviews to outline mobility routines of two older adults. However, their study did not capture the relational dimension through quantitative measures. To fill this gap, we propose a home-centric place network framework that integrates spatial and relational dimensions using quantitative GIS methods.

### *1.1.2 | Home-Centric Place Network With a Dual Perspective*

A home-centric place network adapts the classic ego-centric approach, which is often used to depict social relationships where "alters" (friends or acquaintances) are connected to the "ego" (Scott 2000). In this context, the home functions as the ego, the visited places are the alters, and the frequency of visits indicates connection strength. This network approach foregrounds individual mobility and allows both spatial and relational attributes to be assigned to each place, facilitating compatibility with established mobility measures (Fillekes et al. 2019) and examining links to health outcomes.

In this network, the spatial and relational dimensions of mobility are assigned as place attributes. The spatial dimension includes common mobility measures such as frequency, distance, and diversity of visited places (Fillekes et al. 2019), all of which are associated physical health, cognitive abilities, and overall well-being (Heller et al. 2020; Hirsch et al. 2014; Luo et al. 2023; Moored et al. 2021; Shoval et al. 2010). The relational dimension, drawing on Feld (1981)'s network of social foci, captures the opportunities for social encounters embedded in different types of community places. Variations in the types of places visited shape social environments and the frequency of recognizing familiar faces, fostering mutual recognition, trust, and informal support over time (Small 2009; McFarland et al. 2014; van Dülmen and Klärner 2022).

To assess the opportunities for social encounters, we adopt Browning et al.'s (2014, 2017) ecological network design, which views social opportunities as emerging from the intersection of residents' activity spaces. When individual activity spaces are aggregated, spatial configurations can shape broader social outcomes, such as community attachment and behavioral risks (Browning et al. 2017; Browning and Soller 2014). Building on this approach, we operationalize the opportunities for social encounters as the extent to which neighbors' routine visits overlap across community places. Older adults are more likely to meet neighbors when they visit places that their neighbors also frequent (Gardner 2011). Thus, older adults' routine movements collectively indicate the strength of these potential social opportunities.

Together, the home-centric place network provides a synthetic, scalable framework for capturing both spatial and relational dimension of community mobility. The spatial dimension captures how often, how far, and how diverse older adults' movement is, while the relational dimension reflects the social encounters embedded in the places they routinely visit. This dual perspective better illuminates how everyday mobility and social interactions jointly shape mental health and well-being.

### *1.1.3 | Aging in Shanghai and Under the COVID-19 Pandemic*

Shanghai has the highest proportion of older adults in China. In 2024, it had 5.8 million residents aged 60+, accounting for 37.6% of its population (Shanghai Civil Affairs Bureau 2025). In response to this demographic shift, the Shanghai government launched the "9073" care infrastructure in 2005, which aims to allocate 90% of elderly care to home-based, community-supported services (Chen and Han 2016). The city also invested heavily in building community care infrastructure. It launched a "15-min community living circle" initiative in 2023, designed to provide essential services within walking distance, as well as medical, recreational, and green spaces (Shanghai Planning and Natural Resources Bureau 2023). Overall, Shanghai's aging care infrastructure prioritizes accessibility of services for older adults at the community level.

The COVID-19 pandemic is a unique crisis that drastically limited physical mobility and everyday social interactions. Unlike many natural disasters, where community networks are essential for immediate aid and recovery (Hawkins and Maurer 2010), COVID-19's spatial spread led to deliberate avoidance of movement and face-to-face contact. Government-enforced social distancing policies further constrained mobility (Gibbs et al. 2020) and diminished opportunities for people to meet, connect, and maintain community connections (Jaspal and Breakwell 2020).

Shanghai's social distancing measures during the pandemic severely restricted community mobility, particularly during the



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major April-June 2022 outbreak and in the subsequent months through December. During the 2-month major outbreak, the city enforced a full lockdown, with residents staying home and all non-essential services closed (Clennentt and Yiu 2022). After the lockdown eased, many community spaces remained closed, dining out was banned, and only essential services operated. A daily health-check system was introduced, and a QR code indicating infection status was required to access public venues (Santos 2022). These measures have heavily limited mobility (e.g., Xiao et al. 2024), especially for older adults (Liu et al. 2021).

Empirical evidence indicates that disruptions to community life disproportionately impact older adults. Research shows that older adults experienced a pronounced contraction in their daily mobility patterns, often limiting themselves to their immediate neighborhoods or even their own homes (Choe et al. 2022). Although the general population experienced declines in in-person contact, older adults faced even more significant disruptions to their support networks (Asante et al. 2021). Moreover, due to their heightened vulnerability to infection, many older adults voluntarily limited in-person interactions to safeguard themselves and their families (Hu and Qian 2022). Consequently, the pandemic brought additional social and emotional challenges for older adults, who reported higher levels of loneliness and anxiety (Litwin and Levinsky 2022).

Given the pivotal role of community care in Shanghai and the pandemic's disruptions to community mobility, it is crucial to analyze how such a crisis reshaped older adults' community mobility from both the spatial and relational dimensions. While evidence shows that older adults experience pronounced spatial, social, and emotional challenges during the pandemic, systematic research remains limited, especially regarding detailed, longitudinal, community-scale mobility patterns. To our knowledge, Ho et al. (2023) provides one of the few in-depth ethnographic accounts in Asia, showing that social distancing policies not only restricted older adults' mobility but also constrained the support networks they rely on to meet their social and emotional needs. This study, therefore, provides a systematic assessment of the pandemic's impact on community mobility by tracing changes in mobility patterns over time and examining how these shifts relate to older adults' mental health.

## 1.2 | Methods and Data

### *1.2.1* | *GPS Data*

The GPS data were collected using the LBSLab application, a location-based services (LBS) research platform built and maintained in the laboratory of one of the authors of this study. This mini-app, available on WeChat, is one of the many mobile apps developed to assess individual mobility (Giannouli et al. 2022). Users with an individual WeChat account only need to scan a QR code to install and run the program; there is no need to download, install, or manage a separate app.

The LBSLab mini-app enables analyses of GIS data while respecting and protecting users' privacy. Thus, it does not collect the GIS data unless the participant clicks the "check-in" function (Figure 1). More specifically, it does not automatically track the location of mobile phones unless users consent to the collection of their data. The development of the mini-app has been reviewed and approved by the institute with which the lab is affiliated (Zhang et al. 2023).[1]

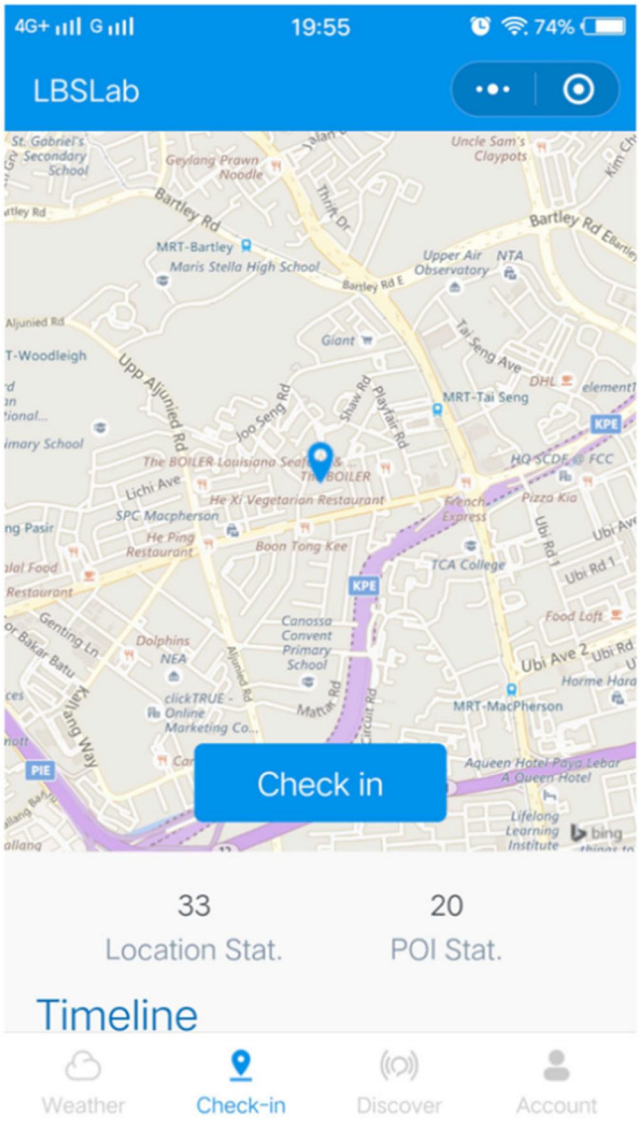



**FIGURE 1** | The "check-in" function in the LBSLab mini-APP on WeChat platform.

When the participant clicks the "check-in" function, a list of nearby points of interest (POIs) appears on the screen from which the participant can choose the most suitable one. The mini-app also records the exact time of check-in, the latitude and longitude coordinates, and the type of place (e.g., commercial venue, public transit, hospital, or school).

### *1.2.2* | *Data Collection*

We collected two waves of GIS data and conducted a wave of survey among participants in two districts in Shanghai, one of

[1]There is no centralized, university-level IRB committee at the lab's home university. Rather, each institute or school reviews and approves the IRB applications submitted by affiliated faculty members.



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China's largest metropolitan areas. One district is in the city center, and the other one is in the suburbs. The participants were asked to report their GPS locations up to seven times per day for 10 days.[2] On these days, the research assistants sent reminders at 8:00, 10:00, 12:00, 14:00, 16:00, 18:00, and 20:00. The first wave was collected in November 2021 and included 174 participants. The second wave was collected in September 2022 and included 85 participants. Our analytical sample includes 64 participants[3] who lived in the surveyed communities, completed both GIS data waves, and completed the survey.

The sharp decrease in the number of participants between the two data waves was due to the major outbreak. 71 of the 85 dropped cases resulted from the residents' committees' refusal. In Shanghai, the residents' committees (*juweihui*) oversee community issues and manage social services for older adults. On most occasions, obtaining their permission to approach older adults is the first step to collecting data. Because the second wave was collected just 3 months after the major outbreak and was still under a strict social distancing policy, some residents' committees refused our request for data collection, citing fears of infection as the main reason.

Statistical analyses revealed that participants not included in the second wave had shorter mobility distances than those who participated in both waves. However, there was no difference in the frequency of visits to out-of-home places (Supporting Information S1: Appendix A). Therefore, the dropped cases tend to be less outgoing than those that remained in the sample. Thus, the reduction in sample size may lead to a biased sample towards more active older adults.

#### 1.2.2.1 | *GIS Data Processing*

ArcGIS (v10.8) was used to process GIS data (i.e., latitude, longitude, and POIs). We simulated mobility paths with the road network downloaded from OpenStreetMap. The Network Analysis tool in ArcGIS was then used to calculate the shortest path from home to each point.

The data cleaning consists of three steps. First, only one point was kept when a participant selected multiple points of the same longitude and latitude within 10 min. This selection was necessary because we found during the pilot phase that users of the mini-app may click a place multiple times within a short period of time to ensure a successful check-in. The rule of 10 min is an arbitrary estimation of the average duration of stay at a point based on the pilot data. Second, points were dropped when the latitude and longitude of a selected point differed significantly from those recorded by the mini-app. Third, points more than 1500 m from the participants' homes are dropped. As mentioned, Shanghai has adopted the concept of "15-min community living circle" to build its age-friendly infrastructure at the community level. Given that individuals over 60 years old tend to walk at a speed of 1.21 m/second (Schimpl et al. 2011) or 4.3 kilometers/hour (Willberg et al. 2023), a 15-min walk covers approximately 1000 m, and a 20-min walk approximately 1500 m. The GIS data (Supporting Information S1: Appendix B) show that 1500 m is a better cutoff point than 1000 m for capturing the daily high-frequency trips of older adults.

#### 1.2.2.2 | *Building the Home-Centric Place Network*

The home-centric place network was built by connecting the older adult's home and all visited points (within 1500 m from home) in a single day. The GIS data do not include the participant's home address. We estimated the home address using longitude and latitude coordinates within the residential community with the highest frequency of check-ins at 20:00. The single-day connections on days of community mobility were then aggregated into a network.[4] Since the outcome is individual-level mental health, 64 home-centric place networks were generated from 1149 days of community mobility, 2496 visited points within 1500 m of home, and 2290 mobility paths.

Figure 2 shows an example of an older adult's home-centric place network, generated from 8 days of community mobility records. Two days were excluded because she traveled long distances (Days 5 and 6). Across these 8 days, four involved movements within 1500 m from home (Days 2, 8–10), three involved movements both within and beyond 1500 m (Days 1, 3, and 7), and one involved movement beyond 1500 m (Day 4). The top panel displays the single-day connections (excluding Days 5 and 6), differentiating between paths within and exceeding 1500 m.

The bottom panel shows the individual-level home-centric place network. Home is at the center; the hollow circles represent visited points, and the lines represent simulated mobility paths. Node size and line thickness represent how often such community mobility paths were repeated.

#### 1.2.2.3 | *Computation and Classification of Two Dimensions*

Table 1 shows the definitions and calculations for the spatial and relational dimensions. The spatial dimension includes three indicators: distance, frequency, and quantity. Distance was measured as the average distance, frequency as the percentage of out-of-home places, and quantity was calculated using Shannon entropy. The relational dimension includes one indicator: sociability. It was calculated as the average sociability rankings of place types.

We used the K-means clustering algorithm to classify the home-centric place network into four clusters. This algorithm is a widely used unsupervised learning method for effectively partitioning a dataset into K distinct clusters, by maximizing the similarity among data points within each cluster and minimizing the similarity between clusters (McQueen 1967). This method has been widely used to classify mobility

[2]We offered a reward of 5 yuan for six check-ins, and participants who completed six daily check-ins over 10 days received an additional 5 yuan.
[3]71 participants were dropped as the result of refusal from residents' committees. Only 18 participants dropped out for personal reasons. Among the remaining 85 participants, 14 did not live in the surveyed communities, and 7 refused to complete the survey.
[4]There are five types of mobility paths: (1) staying at home all day; (2) daily trips within 1500 m from home; (3) daily trips beyond 1500 m from home; (4) daily trips both within and beyond 1500 m from home; (5) long-distance travel (trips not originated from home). Types 3 and 5 were excluded from the analysis. For type 4, only trips within 1500 m to home were included in the analysis.



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**FIGURE 2** | The process of building one participant's home-centric place network from 8 days of mobility paths.

patterns (Gibbs et al. 2020; Hong et al. 2021). The number of clusters was selected based on the silhouette coefficient across candidates with 4–12 clusters, and the four-cluster solution achieved the best silhouette value. Because of the small sample size, we ran robustness checks on alternative clustering methods and on a re-sampling-based stability analysis of the clustering results; both results suggest the stability of four clusters of home-centric place networks.[5]

[5]First, to assess robustness to methodological choices, we re-estimated the clustering using K-medoids and agglomerative clustering and quantified agreement with the K-means solution using normalized mutual information (NMI). The K-medoids solution showed high agreement with the baseline partition (NMI = 0.72), and agglomerative clustering also showed substantial agreement (NMI = 0.63). Second, to mitigate the potential randomness associated with clustering in a small sample, we conducted a resampling-based stability analysis. We performed 500 resampling runs and computed the co-assignment probability matrix. The matrix exhibited a clear block-diagonal structure, and the mean within-cluster co-assignment probability was 0.999, indicating a highly stable partition. Taken together, these results suggest that the identified four-type home-centric place networks are reproducible across alternative clustering methods.



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**TABLE 1** | Description, definition, an calculation of mobility indicators in the home-centric place network.

| Dimension | Indicator | Definition | Calculation |
|---|---|---|---|
| Spatial | Frequency | Percent of out-of-home locations | The number of locations that are outside home divided by the total number of records. |
| | Distance | Mean distance | From home to each visited location weighted by using the number of check-ins in each location in ten days. |
| | Quantity | Shannon entropy of place types | $H = -\sum_{i=1}^{n} p_i \log p_i$ is used to measure the evenness of visit distributions across different location types, where $p_i$ represents the proportion of visits to each location type. Locations are classified in 12 types: home, medical, educational, elderly care, commercial, fitness, public park, neighborhood management, public transit, residents' center, home community, other community. |
| Relational | Sociability | Average ranking in sociability of place types | The sociability of location type is ranked by the number of total check-ins. Home has a sociability ranking of 0. The sociability rankings (from the lowest to the highest) are: old age care center (1), fitness (2), educational (3), medical (4), public transits (5), non-home gated community (6), public park (7), residents' committee (8), community center (9), home gated community (10); commercial (11). $\frac{\sum_{i=1}^{11} r_i * f_i}{11}$ is used to measure the average ranking, where $r_i$ represents the sociability ranking of location type $i$ and $f_i$ represents the frequency of visits of location type $i$. |

### 1.3 | Survey Data

#### *1.3.1* | *Data Collection*

The survey was conducted online in 2022. On the last day of the GIS data collection, we personally sent the questionnaire to participants via WeChat. It included information about older adults' basic demographic and socioeconomic characteristics and mental health. It took about 5–6 min to complete.

### 1.4 | Variables

Mental health was measured with psychological distress, assessed with a modified version of the CESD-10 (Andresen et al. 1994). The scale includes 10 questions, and principal component analysis (PCA) was used to reduce data dimensionality. The analysis yielded two factors, which together explain approximately 74% of the variance. The first factor measures the level of negative emotions, and the second factor measures the level of negative future expectations.[6]

Control variables included gender (1 = male, 0 = female), age, educational level,[7] monthly income,[8] living arrangements (0 = living with someone, 1 = living alone), and city districts (1 = suburb, 0 = city center). These individual characteristics are commonly used in mobility-related aging research (Bevilacqua et al. 2024; Gillespie and Fokkema 2024). Indicators of migration status could not be included in the analyses because all of the participants were Shanghai residents.

### 1.5 | Statistical Methods

Ordinary least squares (OLS) regression was used to examine the impact of home-centric place networks on older adults' mental health (psychological distress). Each older adult has a preference for certain mobility patterns. Therefore, to reduce unobserved heterogeneity, we used the changes in the home-centric place network from 2021 to 2022 as the independent variable. We also controlled for distance in 2021 as a baseline because this is one of the most important indicators of walking accessibility (Willberg et al. 2023).

## 2 | Results

The participants' individual characteristics are presented in Table 2. They were, on average, 67 years old, middle-school educated, and 42% lived in the city center (a design effect). The sample was slanted towards women (84.4%). Therefore, the older adults in our sample tended to be relatively young, healthy, and educated, and they knew how to use smartphones efficiently.

### 2.1 | Diversity in Home-Centric Place Network

Based on variations in the four indicators (i.e., frequency, distance, quantity, and sociability), the older adults formed four clusters of place networks that we classified into two pairs: "outgoing" or "homebody" and "adventurous" or "locale-anchored." All four clusters were found among residents in

[6]The first factor measured the level of negative emotions, including seven statements: "I got scared for no reason," "I felt afraid," "I got headaches and dizziness and can't get up for anything," "I blamed myself a lot," "I got nervous," "I didn't sleep well," and "I got depressed." The second indicator measured the level of negative future expectations: "I felt worthless," "I found it difficult to do anything," and "I felt that the future was hopeless."
[7]The levels of education was coded as: 1 = no education/primary school, 2 = junior high/senior high/vocational school; 3 = associate's degree/bachelor's degree or higher.
[8]The level of income was coded as: 1 = 0–2000 yuan, 2 = 2001–5000 yuan, 3 = 5001–8000 yuan, 4 = 8001–15,000 yuan, 5 = 15,001 yuan and over.



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**TABLE 2** | Mean, standard deviation (SD) and range for the mobility indicators, older adults' wellbeing and other individual characteristics per respondent, aggregate over two survey waves ($N = 64$).

| Variables | Mean | SD | Range |
|---|---|---|---|
| Mobility indicators | | | |
| Frequency (%) | 32.4 | 22.9 | 0 to 100 |
| Distance (meters) | 480.012 | 200.704 | 0 to 1314.497 |
| Quantity | 1.058 | 0.601 | 0 to 2.333 |
| Sociability | 24.080 | 23.556 | 0 to 127.818 |
| Psychological distress | | | |
| Negative emotions | 0 | 1 | −1.627 to 2.758 |
| Negative future expectation | 0 | 1 | −1.480 to 3.747 |
| Control variables | | | |
| Male (vs. female) | 0.156 | | 0 to 1 |
| Age | 67.141 | 6.248 | 60 to 76 |
| Monthly income | 2.25 | 0.563 | 1 to 4 |
| Education level | 3.766 | 0.750 | 2 to 5 |
| Live alone | 0.078 | | 0 to 1 |
| City center (vs. suburb) | 0.422 | | 0 to 1 |

both the city center and the suburbs. Figure 3 shows one example of each cluster.

In Figure 3, R(a) represents an outgoing person. He or she visited more out-of-home places (frequency = 73%), visited a variety of places (quantity = 1.95), and preferred places with sociable opportunities (sociability = 41.73). By contrast, R(b) represents a homebody. He or she mostly stayed at home (frequency = 9%), visited fairly similar places (quantity = 0.46), and did not often visit sociable places (sociability = 8.27). This pair differs in frequency, quantity, and sociability.

R(c) represents an adventurous person. They walked farther away from home (distance = 661 m) and often visited sociable places (sociability = 270.91). By contrast, R(d) represents a locale-anchored person. Mobility was limited to community places near home (distance = 282 m), and he or she visited sociable places less often (sociability = 22.91). This pair differs in distance and sociability.

## 2.2 | Pandemic Impact

A major COVID-19 outbreak occurred between two waves of data collection, with a large-scale lockdown in place between March and June 2022. Up to the second wave of data collection, serious concerns about infection persisted among the public, and the government continued to require daily virus testing for mobility. The home-centric place networks have shown noticeable changes due to the major outbreak.

Figure 4 shows the changes in the four clusters of place networks between the two waves of data collection. Part (a) shows the percentage of each cluster in 2021 and 2022. The percentage of participants in the outgoing cluster dropped significantly, from 30% in 2021 to 14% in 2022. The adventurous cluster remained fairly stable at about 25% of the sample. The other two groups increased correspondingly. The homebody cluster increased from 27% to 36%, and the local-anchored cluster increased from 19% to 25%. Thus, older adults had to adjust their community mobility during the pandemic, and those who were outgoing suffered the most.

Part (b) shows the transition rate from 2021 to 2022. Again, the members of the outgoing cluster had the lowest percentage remaining in the same cluster (42%). The rest of the participants either moved to the adventurous (26%) or the locale-anchored cluster (21%). By contrast, the homebody cluster had the highest percentage of members remaining in the same group (76%). The seemingly unchanged percentage of the adventurous cluster reflects both transitions: members of the adventurous cluster moving out to the homebody cluster, and the outgoing cluster moving in to the adventurous cluster. The homebody cluster was also the largest receiving cluster in 2022, followed by the locale-anchored cluster. The sharp rise in the homebody cluster indicates that older adults adapted to a homestay after the outbreak. The rise in the locale-anchored cluster suggests that older adults who left their homes visited only nearby places. These changes are consistent with the consequences of the fear of infection and government-mandated mobility restrictions reported in other research (Dávid et al. 2023; Ho et al. 2023).

To further break down the changes over time, Figure 5 shows the changes in the four indicators between 2021 and 2022. Again, the pattern is similar between those living in the city center and those living in the suburbs. Two indicators, frequency and sociability, showed a significant decrease. Frequency decreased from 37% to 28% ($p = 0.020$), and sociability decreased from 28.91 to 19.26 ($p = 0.018$). Distance and quantity did not show statistical differences. Again, this finding confirms that older adults remained inside and reduced going out for social purposes during the pandemic.

## 2.3 | Older Adults' Mental Health

Table 3 presents three sets of regression results examining the association between the changes in home-centric place network and older adults' psychological distress. Set A used 2022 cluster classifications, set B used changes in clusters, and set C used changes in four indicators. All models include control variables, but none of them are associated with the outcome variables.[9]

Only changes in four indicators (set C) were significantly associated with psychological distress among older adults. None of the coefficients in sets A and B reached statistical

[9]Results available upon requests. The insignificant associations do not mean that individual characteristics are unimportant for older adults' mental health. Rather, they suggest that the sample homogeneity (as shown in Table 2) prevented us from detecting any meaningful associations.



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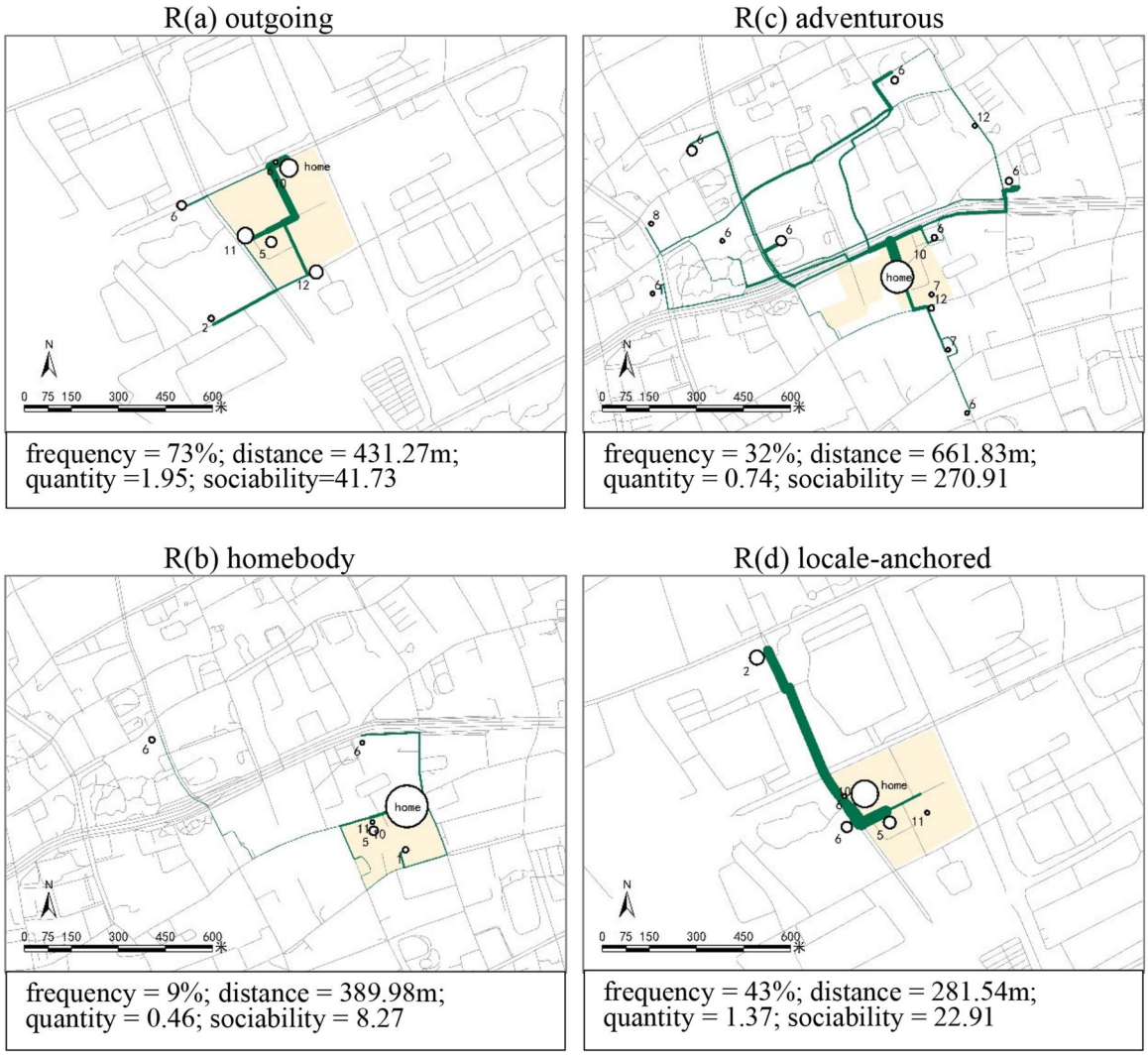



**FIGURE 3** | Four groups of home-centric place networks.

significance. These findings suggested that either older adults preferred certain mobility patterns or the sample size was too small. In set C, four indicators were associated with the level of negative future expectations.

Regarding negative future expectations, both spatial and relational dimensions matter. As expected, higher quantities ($b = -0.691$, $p < 0.05$) and out-of-home frequencies ($b = -1.812$, $p < 0.1$) are associated with lower levels of negative future expectations. Unexpectedly, longer distance ($b = 0.001$, $p < 0.05$) and greater sociability ($b = 0.016$, $p < 0.01$) are associated with higher negative future expectations.

The unexpected results contrast with findings from ordinary times, in which longer walking distances and greater sociability usually increase older adults' health outcomes (den Hoed and Russo 2025). However, we speculate that they may be attributable to the pandemic: walking outside the community and visiting social places were associated with heightened fear of infection (Jaspal and Breakwell 2020). To explore this possibility, we ranked the place types by percentage change in proportion of out-of-home visits between 2021 and 2022 (Supporting Information S1: Appendix C). The results show that older adults chose not to visit common sociable places, including community centers, public parks, or residents' committees.

## 3 | Discussion

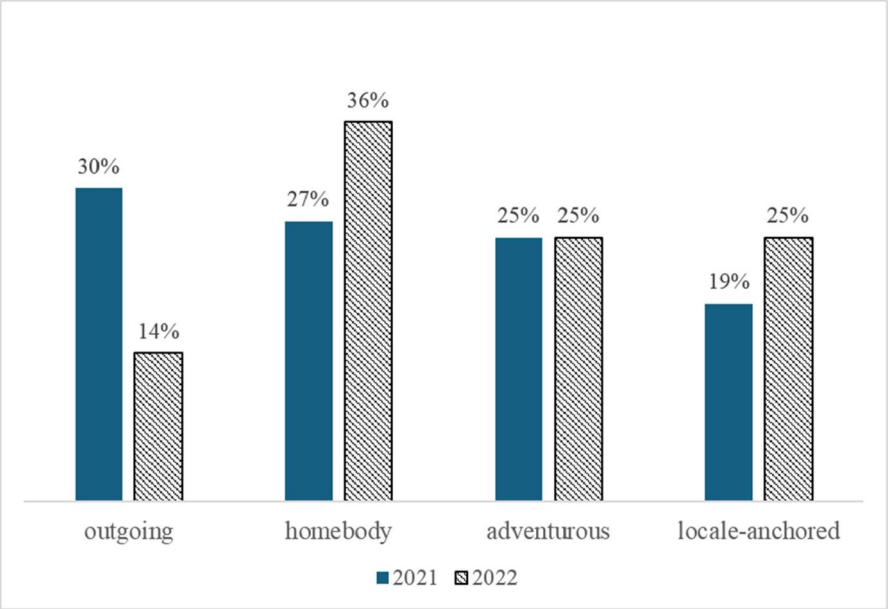
For older adults, community mobility encompasses both spatial and relational dimensions. To address this dual perspective, this study develops a home-centric place network framework and uses a quantitative GIS approach to examine mobility patterns among a sample of older adults in Shanghai during COVID-19 pandemic. Here are a summary of key findings taking from this dual perspective.

First, older adults exhibit diverse mobility patterns, jointly determined by spatial and relational dimensions. They either go outside home very often and visit sociable places (outgoing *vs.* homebody), or walk long distances and visit sociable places (adventurous vs. locale-anchored). Note that this diversity in community mobility is generated from a relatively homogeneous group of older adults. Thus, integrating both spatial and relational dimensions characterizes older adults' different aging experiences in community places.



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(a) percentage from 2021 to 2022




(b) transition rate from 2021 to 2022

| | 2022 | | | |
|---|---|---|---|---|
| 2021 | outgoing | homebody | adventurous | locale-anchored |
| outgoing | 42.11% | 10.53% | 26.32% | 21.05% |
| homebody | -- | 74.47% | 11.76% | 11.76% |
| adventurous | -- | 37.50% | 43.75% | 18.75% |
| locale-anchored | 8.33% | 16.67% | 16.78% | 58.33% |

**FIGURE 4** | Changes in groups of home-centric place networks, from 2021 to 2022 (after a major COVID-19 outbreak in Shanghai).

Second, the pandemic has disrupted both the spatial and relational dimensions of community mobility and produced new forms of inequality. After the outbreak, there was a sharp, significant decline in the frequency of out-of-home visits and sociability, suggesting that older adults spent more time at home and had fewer opportunities for social encounters. A closer look revealed that the decline is more pronounced in the outgoing and adventurous groups than in the other two, suggesting that older adults who were more mobile and sociable before the outbreak suffered more than those who were less so.

Third, the pandemic alters the moral and affective meaning of community mobility. Regression results find increased sociability was associated with more negative expectations about the future. These results align with findings that fear of infection led to social avoidance (Dávid et al. 2023; Steijvers et al. 2022) but stand in contrast to evidence from non-crisis periods, which typically associates social interactions in community places with better well-being in later life (Zhang and Huang 2023; Song et al. 2023). Thus, a home-centric place network helps clarify the fragile nature of community mobility, in which social practices that normally foster connection can easily be replaced by risk-averse withdrawal.

To summarize, we aim to reconceptualize mobility as participation in relationally embedded spatial networks. This approach recognizes that older adults' movements are not just patterns of exposure to locations, but are shaped by, and contribute to, the social relationships and opportunities for interaction found in those places. We believe this dual perspective will reframe the concept of mobility and offer a deeper, more nuanced understanding of older adults' experiences in community life.

There are several limitations in the quantitative analyses that future research could build on. First, our spatial measures did not capture the temporal scale of mobility, as the check-in timestamps were too sparse to reveal meaningful temporal patterns. Second, our sociability measure estimated opportunities for social encounters but did not differentiate the meanings in those social interactions. Gardner (2011), for example, distinguished between community "third places" (e.g., parks and cafés) that foster causal conversation and "transitory zones" (e.g., sidewalks and grocery store lines) with more passive encounters, but our small sample size prevented further exploration of these patterns. Finally, our regression results should be viewed as suggestive because the sample lacks representation and small sample size raising concerns about statistical power. Thus, we view this dataset's primary value by documenting diversity and changes of older adults' pattern of community mobility during the pandemic, serving as a demonstration of the dual perspective and home-centric place network.

## 4 | Conclusion

This study advances the field by moving beyond traditional notions of "spatial exposure" to conceptualize mobility as participation in relationally embedded spatial networks. Previous



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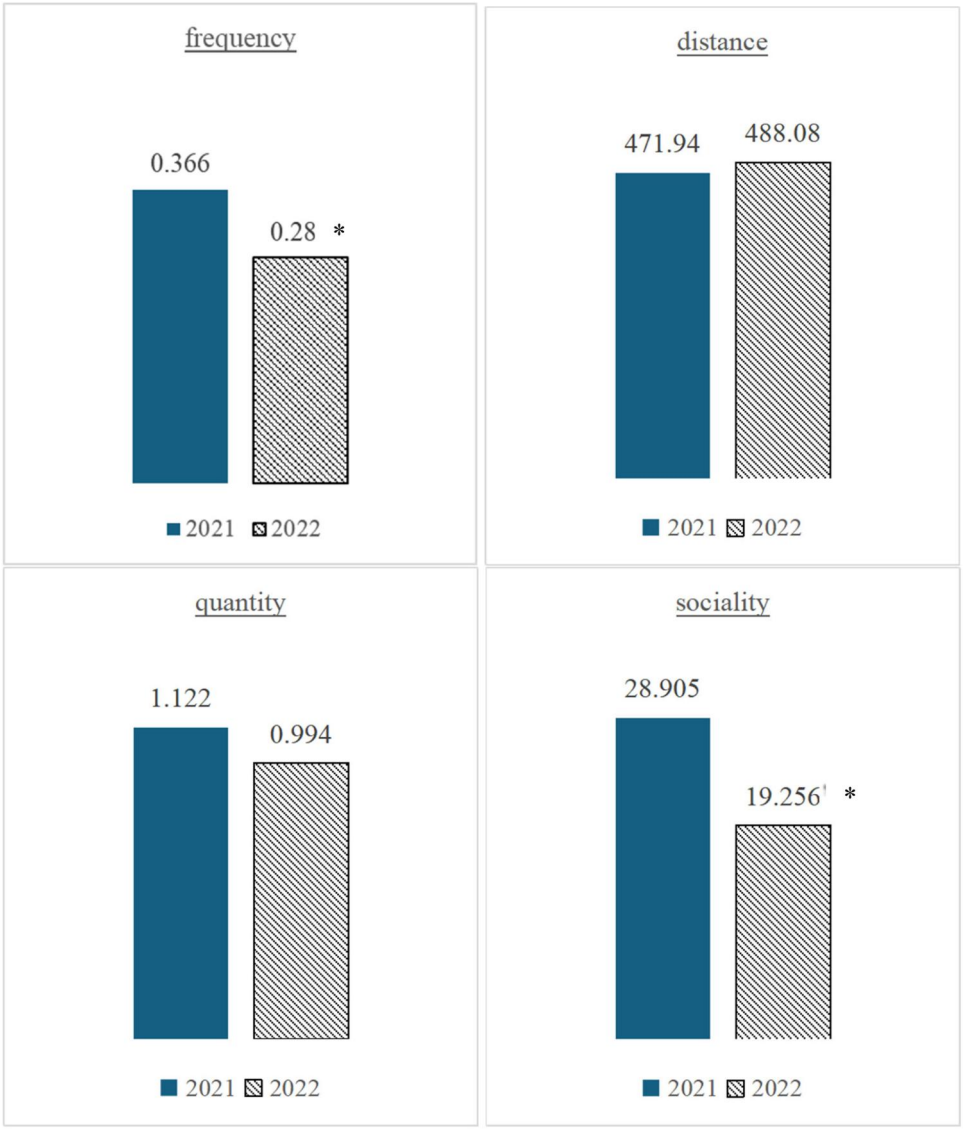



**FIGURE 5** | Changes in mobility indicators, from 2021 to 2022 (after a major COVID-19 outbreak in Shanghai).

**TABLE 3** | Regression analyses to examine associations between the changes in home-centric place network and older adults' mental health (psychological distress) ($N = 64$).

| | Negative emotions | Negative future expectations |
|---|---|---|
| A: cluster in 2022 (reference = outgoing) | | |
| Homebody | 0.412 (0.448) | 0.490 (0.471) |
| Adventurous | 0.372 (0.435) | 0.288 (0.457) |
| Local-anchored | 0.699 (0.435) | 0.558 (0.458) |
| B: changes in clutsers (reference = other) | | |
| Into homebody cluster | 0.041 (0.378) | 0.367 (0.393) |
| Out of outgoing cluster | −0.290 (0.379) | 0.255 (0.399) |
| C: changes in four indicators | | |
| Frequency (%) | 0.073 (1.065) | −1.812$^{+}$ (1.028) |
| Distance | 0.004 (0.001) | 0.001* (0.0005) |
| Quantity | 0.360 (0.297) | −0.576* (0.287) |
| Sociability | 0.002 (0.007) | 0.016** (0.007) |

*Note:* Control variables include male, age, income level, education level, live alone, center (vs. suburb), perceived no conflict and frequency in 2021(%). Standard error in parathesis.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.
$^{+}p < 0.1$.



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research has often examined spatial and relational dimensions of aging separately, limiting our understanding of how their interplay shapes older adults' mental health. By synthesizing these perspectives through the concept of "aging in networks," and operationalizing them via a home-centric place network, our approach resolves the problem of parallel literatures and reveals how everyday mobility is fundamentally shaped by both the places visited and the opportunities for social connection they afford.

The introduction of a sociability measure enables us to quantify the opportunities for social encounters embedded within mobility patterns. This is a key advancement over prior GIS-based approaches that focused solely on movement metrics. By capturing the intersection of where and with whom older adults are likely to interact, this advancement allows for a more nuanced analysis of how community mobility supports or undermines social connection and mental health.

Our findings during the COVID-19 pandemic further demonstrate that community mobility is not just about accessing places, but about maintaining social embeddedness in times of crisis. This reframing shows that the fragility or resilience of older adults' everyday lives is best understood through the dynamic interplay of spatial and relational dimensions, offering new theoretical and practical insights into how mobility supports or undermines mental health and social inclusion.

These insights carry important implications for policies aimed at strengthening community resilience. If community mobility can shift so quickly from a source of support to a site of risk, interventions need to focus not only on restoring movement after crises but also on making everyday mobility practices more resilient to disruption. This includes designing public spaces that enable flexible, low-risk forms of social interaction (e.g., modular gathering designs) and developing clear, trusted communication channels that help older adults assess and manage risk without retreating entirely from communal life. By recognizing the vulnerability of community mobility and planning for its reconfiguration during crises, policymakers can help ensure that communal places and routine activities in them remain sources of resilience rather than anxiety, both in crisis and non-crisis periods.


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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section.
**Supporting File:** 3_psp70389-sup-0001-Appendices-RR-yc_19oct.docx.



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